

Applied Machine Learning Lab

Experiment 01

Python ML Libraries: Practical Skills for Data Analysis

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AIM:

To understand and use Python ML libraries like NumPy, Pandas and Matplotlib for data analysis.

THEORY:

Python libraries like NumPy, Pandas and Matplotlib are widely used for data analysis, data manipulation and visualization.

Commands/Code Used:

```
from __future__ import annotations
```

```
import statistics
```

```
from pathlib import Path
```

```
import matplotlib
```

```
matplotlib.use("Agg")
```

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
import pandas as pd
```

```
from scipy import linalg
```

```
def numpy_demo():
```

```
    arr_1d = np.arange(1, 13)
```

```
    arr_2d = arr_1d.reshape(3, 4)
```

```
    arr_3d = arr_1d.reshape(2, 2, 3)
```

```
    print(arr_1d)
```

```
    print(arr_2d)
```

```
    print(arr_3d)
```

```
def pandas_demo():
```

```
    series_a = pd.Series([10, 20, 30], index=["a", "b", "c"])
```

```
    series_b = pd.Series([2, 4, 5], index=["a", "b", "c"])
```

```
print(series_a / series_b)
```

```
def matplotlib_demo():
```

```
    x = [1, 2, 3, 4, 5]
```

```
    y = [2, 4, 1, 8, 7]
```

```
    plt.figure()
```

```
    plt.plot(x, y, color="red", marker="o")
```

```
    plt.title("Red_Line_Graph")
```

```
    plt.xlabel("X-axis")
```

```
    plt.ylabel("Y-axis")
```

```
    plt.grid(True)
```

```
    plt.savefig("exp_1_red_line_plot.png")
```

```
    plt.close()
```

```
def scipy_demo():
```

```
    matrix = np.array([[1, 2], [3, 4]])
```

```
    print(linalg.det(matrix))
```

```
    print(linalg.eig(matrix))
```

```
def statistics_demo():
```

```
    data = [2, 4, 4, 4, 5, 5, 7, 9]
```

```
    print(statistics.mean(data))
```

```
print(statistics.median(data))  
print(statistics.mode(data))
```

```
def main():  
    numpy_demo()  
    pandas_demo()  
    matplotlib_demo()  
    scipy_demo()  
    statistics_demo()
```

```
if __name__ == "__main__":
```

```
    main()RESULTS:
```

Successfully performed basic data operations and visualization.

CONCLUSION:

The experiment helped in understanding basic Python ML libraries.